

Assignment 2

Statistical Inference 2026-27

October 8, 2026

Discussion among yourselves is permitted and encouraged. AI usage is permitted.

Problem 1: Better estimation, better testing? (10 points)

Suppose $X \sim N_p(\theta, I_p)$, where $p \geq 3$ and $\theta \in \mathbb{R}^p$. The maximum likelihood estimator $\hat{\theta} = X$ is unbiased. Since the Fisher information matrix is I_p , the multivariate Cramér–Rao inequality implies that every unbiased estimator $T(X)$ of θ , with finite second moments, satisfies

$$\text{Cov}_\theta(T) \succeq I_p,$$

where $A \succeq B$ means that $A - B$ is positive semidefinite. Consequently, under squared-error loss,

$$R(\theta, T) := E_\theta[\|T(X) - \theta\|^2] = \text{tr}(\text{Cov}_\theta(T)) \geq p.$$

The estimator X attains this bound for every θ , so it has the smallest risk among unbiased estimators.

Allowing bias changes the conclusion. The **James–Stein estimator** is

$$\hat{\theta}_{\text{JS}} = \left(1 - \frac{p-2}{\|X\|^2}\right) X.$$

You may take as given that

$$R(\theta, \hat{\theta}_{\text{JS}}) = p - (p-2)^2 E_\theta \left[\frac{1}{\|X\|^2} \right] < p = R(\theta, X) \quad \text{for every } \theta.$$

Thus, X attains the Cramér–Rao bound among unbiased estimators yet is inadmissible when biased estimators are allowed. This does not contradict the bound, since the James–Stein estimator is biased. The improvement concerns the sum of the coordinate mean squared errors, not necessarily each coordinate separately.

When $\|X\|^2 < p-2$, the James–Stein shrinkage factor is negative. The **positive-part James–Stein estimator** avoids reversing the direction of X :

$$\hat{\theta}_{\text{JS}}^+ = \left(1 - \frac{p-2}{\|X\|^2}\right)_+ X, \quad (u)_+ := \max\{u, 0\}.$$

Define both estimators to be zero at $X = 0$. You may also take as given that the positive-part estimator has risk no greater than the original James–Stein estimator for every θ .

Does this improvement in estimation translate into better hypothesis tests? Consider

$$H_0 : \theta = 0 \quad \text{against} \quad H_1 : \theta \neq 0,$$

at significance level $0 < \alpha < 1$.

a. Does better estimation give a better test? (4 points)

A student argues:

“Since the James–Stein estimator improves on X , a test rejecting for large $\|\hat{\theta}_{\text{JS}}^+\|$ should be more powerful than a test rejecting for large $\|X\|$.”

Derive the level- α likelihood-ratio test. Then calibrate the proposed James–Stein-based test to the same level and compare their rejection regions. Is the student’s argument correct?

Would the rejection regions still coincide if the original James–Stein estimator were used instead of its positive-part version? Justify your answer by examining its norm; you need not calculate the new critical value.

Hint: Express $\|\hat{\theta}_{\text{JS}}^+\|$ as a function of $r = \|X\|$. You may assume that $\chi_{p,1-\alpha}^2 > p - 2$, where $\chi_{p,1-\alpha}^2$ denotes the $(1 - \alpha)$ -quantile of the chi-square distribution with p degrees of freedom.

b. Does restricting attention to unbiased tests yield a UMPU test? (4 points)

Consider the coordinate tests

$$\phi_j(X) = \mathbf{1}\{|X_j| > z_{1-\alpha/2}\}, \quad j = 1, 2,$$

where z_q is the q -quantile of the standard normal distribution.

Show that both are unbiased level- α tests for the vector hypothesis above. Use the following hint and the Neyman–Pearson lemma to show that no UMPU test exists. Explain precisely where you use the uniqueness, up to null sets, of the most powerful test, and why that uniqueness holds here. Explain briefly why the nonexistence of a UMPU test does not contradict James–Stein risk improvement.

Hint: Let f_θ denote the density of $N_p(\theta, I_p)$, and let e_j be the j th coordinate vector. For any $a > 0$, the equal-weight mixture of the alternatives $\theta = ae_j$ and $\theta = -ae_j$ has likelihood ratio

$$\frac{\frac{1}{2}f_{ae_j}(x) + \frac{1}{2}f_{-ae_j}(x)}{f_0(x)} = e^{-a^2/2} \cosh(ax_j).$$

c. Explain your reasoning. (2 points) If you used AI, quote one step from its solution that you initially found unclear. If you did not use AI, identify one step in your own solution that required the most thought.

Explain that step in your own words, including why it is valid. Do not simply repeat the calculation or name a theorem.

Problem 2: NHANES data analysis (15 points)

A health analyst claims:

“Physically active adults have lower systolic blood pressure. If two different tests are significant, this establishes that physical activity reduces blood pressure.”

Investigate this claim using `NHANES::NHANES` in R. Use two-sided tests at the 5% level, and restrict the analysis to adults aged 20-69 with observed `PhysActive` and `BPSysAve`. These variables record reported recreational physical activity and average systolic blood pressure, respectively.

The package provides health-survey data adapted and resampled for teaching. For this problem, treat records as independent observations under a simple-random-sample approximation. Discuss the limits of this convention when interpreting the results; the exercise does not support design-valid inference for the U.S. population. Consult the [package documentation](#).

- a. Define the question and examine the data. (2 points)** State the parameter of interest and hypotheses mathematically. Report exclusions, group sizes, and suitable summaries. Include one plot and identify features that matter for inference.
- b. Compare the groups. (4 points)** Estimate

$$\Delta = E[Y \mid A = \text{Yes}] - E[Y \mid A = \text{No}],$$

where Y is systolic blood pressure and A is activity status. Choose and justify a testing procedure, and report the estimated difference, a 95% confidence interval, and a p -value. Use a second approach involving bootstrap or permutation methods to assess the robustness of your inference. Describe its implementation and assumptions. Explain whether the two approaches address the same null hypothesis and how any agreement or disagreement should be interpreted.

- c. Investigate age as an alternative explanation. (3 points)** Compare the age distributions of the activity groups. Repeat the mean comparison within the age bands 20–39, 40–59, and 60–69, reporting estimates and confidence intervals. Compare these results with the pooled analysis. What do they suggest about confounding, and what remains unresolved? Treat the subgroup comparisons as exploratory.
- d. Stress-test an assumption. (3 points)** Construct two synthetic populations with equal means but unequal variances, using the observed group sizes and approximate variance ratio as guides. Compare the rejection rates of two testing procedures of your choice over at least 500 simulated datasets. Specify the distributions and report Monte Carlo standard errors. Explain what the results reveal about the procedures' assumptions and what they cannot establish about the NHANES analysis. Your simulated null must preserve the unequal variances.
- e. Evaluate the claim and audit your reasoning. (3 points)** Write a verdict addressing the analyst's claim, effect magnitude, uncertainty, missing data, and the sampling convention. Document one consequential AI recommendation you accepted, modified, or rejected, and explain how you checked it. If you did not use AI, give a plausible methodological recommendation instead in this data analysis.

Submission. Submit a short report and a reproducible R/Python script, including random seeds. DO NOT print and submit your code. Instead upload it on onedrive in [this link](#) as `A2.Firstname.R/A2.Firstname.py`.